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Untitled0.ipynb

[11] ✓ 0s

```
jug4 = 0
jug3 = 0

print("Initial State:", (jug4, jug3))

# Step 1: Fill the 3-gallon jug
jug3 = 3
print("Fill 3-gallon jug:", (jug4, jug3))

# Step 2: Pour water into the 4-gallon jug
jug4 = jug4 + jug3
jug3 = 0
print("Pour into 4-gallon jug:", (jug4, jug3))

# Step 3: Fill the 3-gallon jug again
jug3 = 3
print("Fill 3-gallon jug:", (jug4, jug3))

# Step 4: Pour until the 4-gallon jug is full
transfer = min(4 - jug4, jug3)
jug4 = jug4 + transfer
jug3 = jug3 - transfer
print("Pour water:", (jug4, jug3))

# Step 5: Empty the 4-gallon jug
jug4 = 0
print("Empty 4-gallon jug:", (jug4, jug3))

# Step 6: Pour the remaining water into the 4-gallon jug
jug4 = jug3
jug3 = 0
print("Final State:", (jug4, jug3))

print("Water in 4-gallon jug:", jug4, "gallons")
```

```
Initial State: (0, 0)
Fill 3-gallon jug: (0, 3)
Pour into 4-gallon jug: (3, 0)
```


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[12]
✓ 0s

```python
class MarsRover:
    def __init__(self):
        self.position = 0
        self.battery = 100
        self.samples = []

    def perceive(self, terrain, sample):
        print("Terrain:", terrain)
        print("Sample detected:", sample)

    def move(self):
        if self.battery >= 10:
            self.position += 1
            self.battery -= 10
            print("Rover moved to position:", self.position)
        else:
            print("Low battery! Rover cannot move.")

    def collect_sample(self, sample):
        if sample:
            self.samples.append(sample)
            print("Sample collected:", sample)
        else:
            print("No sample available.")

    def analyze_sample(self):
        if self.samples:
            print("Analyzing sample:", self.samples[-1])
            print("Sample analysis completed.")
        else:
            print("No samples to analyze.")

    def status(self):
        print("Position:", self.position)
        print("Battery:", self.battery, "%")
        print("Samples collected:", len(self.samples))


rover = MarsRover()

rover.perceive("Rocky", "Rock")
```

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[12] ✓ 0s

```
rover.move()
rover.collect_sample("Rock")
rover.analyze_sample()
rover.status()
```

```
Terrain: Rocky
Sample detected: Rock
Rover moved to position: 1
Sample collected: Rock
Analyzing sample: Rock
Sample analysis completed.
Position: 1
Battery: 90 %
Samples collected: 1
```

[13] ✓ 0s

```
N = 8
board = [-1] * N

def is_safe(row, col):
    for previous_col in range(col):
        previous_row = board[previous_col]

        if previous_row == row:
            return False

        if abs(previous_row - row) == abs(previous_col - col):
            return False

    return True

def solve_queens(col):
    if col == N:
        return True

    for row in range(N):
        if is_safe(row, col):
            board[col] = row

            if solve_queens(col + 1):
                return True
```

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[13] ✓ 0s

```
            board[col] = -1

    return False

if solve_queens(0):
    print("Solution for 8 Queens Problem:")
    for row in range(N):
        for col in range(N):
            if board[col] == row:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()

    print("\nQueen positions (row, column):")
    for col in range(N):
        print((board[col] + 1, col + 1))
else:
    print("No solution found.")
```

```
Solution for 8 Queens Problem:
Q . . . . . . .
. . . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .

Queen positions (row, column):
(1, 1)
(5, 2)
(8, 3)
(6, 4)
(3, 5)
(7, 6)
(2, 7)
(4, 8)
```

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[14] 1m

```
def ola_cab_booking():
    print("Welcome to OLA Cab Booking")

    source = input("Enter starting location: ")
    destination = input("Enter destination: ")

    cabs = {
        "Mini": 100,
        "Micro": 80,
        "Sedan": 150,
        "Shared": 60,
        "Prime": 200
    }

    print("\nAvailable Cab Types:")
    for cab, fare in cabs.items():
        print(cab, "- Rs.", fare)

    choice = input("\nEnter your preferred cab: ").title()

    if choice in cabs:
        print("\nBooking Details")
        print("Starting Location:", source)
        print("Destination:", destination)
        print("Selected Cab:", choice)
        print("Estimated Fare: Rs.", cabs[choice])

        confirm = input("Confirm booking? (yes/no): ").lower()

        if confirm == "yes":
            print("Cab booked successfully!")
            print("Your cab is on the way.")
        else:
            print("Booking cancelled.")
    else:
        print("Invalid cab type.")

ola_cab_booking()
```

```
Welcome to OLA Cab Booking
```

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```
Welcome to OLA Cab Booking
Enter starting location: salem
Enter destination: chennai

Available Cab Types:
Mini - Rs. 100
Micro - Rs. 80
Sedan - Rs. 150
Shared - Rs. 60
Prime - Rs. 200

Enter your preferred cab: mini

Booking Details
Starting Location: salem
Destination: chennai
Selected Cab: Mini
Estimated Fare: Rs. 100
Confirm booking? (yes/no): yes
Cab booked successfully!
Your cab is on the way.
```

Double-click (or enter) to edit

[15]
0s

```python
import heapq

def uniform_cost_search(graph, start, goal):
    priority_queue = [(0, start, [start])]
    visited = set()

    while priority_queue:
        cost, node, path = heapq.heappop(priority_queue)

        if node in visited:
            continue

        visited.add(node)

        print("Visiting:", node, "| Cost:", cost)

        if node == goal:
```

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[15] ✓ 0s

```
            print("Total cost:", cost)
            return path, cost

        for neighbor, edge_cost in graph.get(node, []):
            if neighbor not in visited:
                new_cost = cost + edge_cost
                new_path = path + [neighbor]

                heapq.heappush(
                    priority_queue,
                    (new_cost, neighbor, new_path)
                )

    print("No path found.")
    return None, float("inf")


# Example weighted graph
graph = {
    'S': [('A', 2), ('B', 5)],
    'A': [('B', 1), ('G', 7)],
    'B': [('G', 3)],
    'G': []
}

uniform_cost_search(graph, 'S', 'G')
```

```
Visiting: S | Cost: 0
Visiting: A | Cost: 2
Visiting: B | Cost: 3
Visiting: G | Cost: 6

Least-cost path: S -> A -> B -> G
Total cost: 6
(['S', 'A', 'B', 'G'], 6)
```

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```python
from collections import deque

GOAL = (1, 2, 3, 4, 5, 6, 7, 8, 0)  # 0 = blank

# Positions the blank can swap with, for each blank index
NEIGHBORS = {
    0: (1, 3),    1: (0, 2, 4),    2: (1, 5),
    3: (0, 4, 6), 4: (1, 3, 5, 7), 5: (2, 4, 8),
    6: (3, 7),    7: (4, 6, 8),    8: (5, 7),
}

def is_solvable(state):
    tiles = [x for x in state if x]
    inv = sum(a > b for i, a in enumerate(tiles) for b in tiles[i + 1:])
    return inv % 2 == 0

def solve(start):
    queue = deque([(start, [start])])
    visited = {start}
    while queue:
        state, path = queue.popleft()
        if state == GOAL:
            return path
        z = state.index(0)
        for n in NEIGHBORS[z]:
            s = list(state)
            s[z], s[n] = s[n], s[z]
            s = tuple(s)
            if s not in visited:
                visited.add(s)
                queue.append((s, path + [s]))

def show(state):
    for i in range(0, 9, 3):
        print(*(x or "_" for x in state[i:i + 3]))
    print()

if __name__ == "__main__":
    raw = input("Enter start state (0 = blank), e.g. 123405678: ").replace(" ", "")
    start = tuple(int(c) for c in raw)
```

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[2] ✓ 19s

```
raw = input("Enter start state (0 = blank), e.g. 123405678: ").replace(" ", "")
start = tuple(int(c) for c in raw)

if len(start) != 9 or sorted(start) != list(range(9)):
    print("Invalid input: use digits 0-8 exactly once.")
elif not is_solvable(start):
    print("Not solvable.")
else:
    path = solve(start)
    for i, s in enumerate(path):
        print(f"Step {i}:")
        show(s)
    print(f"Solved in {len(path) - 1} moves.")
```

```
Enter start state (0 = blank), e.g. 123405678: 123405678
Step 0:
1 2 3
4 _ 5
6 7 8

Step 1:
1 2 3
4 5 _
6 7 8

Step 2:
1 2 3
4 5 8
6 7 _

Step 3:
1 2 3
4 5 8
6 _ 7

Step 4:
1 2 3
4 5 8
_ 6 7
```

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```
5 _ 8
4 6 7

Step 7:
1 2 3
5 6 8
4 _ 7

Step 8:
1 2 3
5 6 8
4 7 _

Step 9:
1 2 3
5 6 _
4 7 8

Step 10:
1 2 3
5 _ 6
4 7 8

Step 11:
1 2 3
_ 5 6
4 7 8

Step 12:
1 2 3
4 5 6
_ 7 8

Step 13:
1 2 3
4 5 6
7 _ 8

Step 14:
1 2 3
4 5 6
7 8 _

Solved in 14 moves.
```

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```
Step 14:
1 2 3
4 5 6
7 8 _

Solved in 14 moves.
```

[4] ✓ 0s

```
N = 8
solutions = []

def is_safe(board, row, col):
    # board[r] = column of the queen placed in row r
    for r in range(row):
        c = board[r]
        if c == col:                          # same column
            return False
        if abs(c - col) == abs(r - row):  # same diagonal
            return False
    return True

def solve(board, row):
    if row == N:                              # all 8 queens placed
        solutions.append(board[:])
        return
    for col in range(N):
        if is_safe(board, row, col):
            board.append(col)                 # place queen
            solve(board, row + 1)             # go to next row
            board.pop()                       # backtrack (remove queen)

def print_board(board):
    for r in range(N):
        print(" ".join("Q" if board[r] == c else "." for c in range(N)))
    print()

solve([], 0)
print("Total solutions:", len(solutions))
print("\nFirst solution:\n")
print_board(solutions[0])
print("Queen positions (row, col):", [(r, c) for r, c in enumerate(solutions[0])])
```

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```
Total solutions: 92
...

First solution:

Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .

Queen positions (row, col): [(0, 0), (1, 4), (2, 7), (3, 5), (4, 2), (5, 6), (6, 1), (7, 3)]
```